

$\text{sqrt}(1.11_2 \times 2^3) = \text{sqrt}(1.11_2) \times \text{sqrt}(2^3) = \text{sqrt}(448 \times 2^{-8}) \times \text{sqrt}(2^3)$ $1.11000000 \longrightarrow 000001.1101(1)$	$(29 \times 2^{-4}) \times 2^{1.5}$ Fractional exponent
$= \text{sqrt}(11.1_2) \times \text{sqrt}(2^2) = \text{sqrt}(56 \times 2^{-4}) \times \text{sqrt}(2^2)$ $11.1000 \longrightarrow 000001.11(1)$	$(7 \times 2^{-2}) \times 2^1$ Not enough precision for rounding
$= \text{sqrt}(448 \times 2^{-7}) \times \text{sqrt}(2^2)$ $11.1000000 \longrightarrow 000010101(1)$	$(21 \times 2^{-3.5}) \times 2^1$ Fractional units
$= \text{sqrt}(896 \times 2^{-8}) \times \text{sqrt}(2^2)$ $11.10000000 \longrightarrow 000001.1101(1)$	$(29 \times 2^{-4}) \times 2^1$ Correct $= 1.11 \times 2^1$